

ALAD: A New Unsupervised Time Series Anomaly Detection Paradigm Based on Activation Learning

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Abstract—Time series anomaly detection has been received growing interest in industrial and academic communities due to its substantial theoretical value and practical significance in reality. Recent advanced methods for time series anomaly detection are based on deep learning techniques, since they have shown their superiority in some specific situations. However, most existing deep learning-based anomaly detection methods require predefined, specific tasks of reconstruction or prediction, necessitating task-specific loss functions. Designing such anomaly-aware loss functions poses a significant challenge due to the ambiguity in defining ground-truth anomalies. Moreover, these methods often rely on complex network architectures that tend to lead to over-generalization, resulting in even abnormal data being well reconstructed or fitted. To mitigate this situation, grounded in activation learning theory, we propose a novel unsupervised time series anomaly detection paradigm termed ALAD. ALAD utilizes a straightforward fully connected network architecture, measuring the typicality of input patterns through the sum of the squared output. Despite its simplicity, ALAD achieves competitive performance compared to state-of-the-art models trained using back-propagation. By utilizing various real-world and synthetic datasets, experimental results have confirmed the effectiveness and feasibility of the proposed paradigm. This work also demonstrates that biologically-plausible local learning can sometimes outperform backpropagation in real-world scenarios.

Index Terms—Anomaly detection, time series analysis, biologically-plausible learning, unsupervised learning.

I. INTRODUCTION

WITH the rapid developing of modern technologies, data from various fields such as finance, economics and finance are recorded as time series data, and thus there is a growing interest in modeling and exploring time series data. Wherein,

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time series anomaly detection as the classic problem has been widely studied by researchers, which has provided necessary technique guarantees for diversified application scenarios, such as industrial Internet, Internet of Things and distributed computing, etc [1], [2], [3]. Effective anomaly detection in time series data can enhance the robustness, reliability, and performance of various systems and processes by providing insights into underlying patterns and trends. This can inform decision-making, optimize resource allocation, and lead to increased efficiency, reduced costs, and improved overall functionality [4].

Time series anomaly detection is the process of identifying unusual, unexpected patterns or behaviors in time series data. It involves developing models that can automatically detect unusual patterns or events in time series data, which may be caused by various factors such as errors in data collection, changes in the underlying system dynamics, or external disturbances. Most of the existing advanced anomaly detection methods are based on deep learning models, which typically adopt reconstruction or forecasting-based models and perform anomaly detection based on the deviation errors between real values and reconstruction or forecasting results [5]. For reconstruction-based models, the autoencoder architecture is usually applied. For instance, Kieu et al. [6] proposed an RNN-based autoencoder framework for time series anomaly detection, since recurrent neural network had been shown to be effective for time series modeling. Li et al. [7] present a temporal convolutional network autoencoder, and performed anomaly detection based on the reconstruction error. For forecasting-based models, Deng et al. [8] applied a graph neural network as the predictor, and used the deviation between the predicted value and the true value as the anomaly criterion. Wu et al. [9] proposed a long short-term memory (LSTM) based anomaly detection method, which was built on normal time series and detected anomalies by utilizing the predictive error for the Gaussian Naive Bayes model.

Although these deep learning-based models have shown their superiority in some specific situations, they inherently suffer from the limitations associated with the following technique-specific challenges. First, these methods generally assume the models will fail to reconstruct or forecast the anomalous samples, since they are trained only using normal samples. However, this assumption does not always holds in time series data, since these methods mainly focus on the quality of reconstruction or forecasting. This may leads to an over-generalization [10] of model, thus it would cause even abnormal data to be well reconstructed or fitted [11]. Second, these tailored anomaly detection models are typically more complicated than the

conventional neural network models. Furthermore, the core algorithm for training deep learning-based anomaly detection methods is backpropagation (BP), which requires predefining a task-specific loss function between the outputs of network and expectations. However, designing anomaly-aware loss functions poses significant challenges due to the ambiguity of defining ground-truth anomalies [12]. Additionally, the training procedure for these elaborate models demands additional resources for gradient computations and storage, which limits their applicability in resource-limited scenarios. Third, the training data from real-world sources are often unbalanced with limited labeled anomalies, and most of them exhibit diverse or even unseen patterns that are not known a priori. Due to these constraints, how to develop a new paradigm for time series anomaly detection is still an open problem.

In this work, we address the challenging problems of time series anomaly detection by developing a simple yet promising architecture termed ALAD (Activation Learning based Anomaly Detection). It does not rely on complicated network architectures or any additional mechanisms, such as contrastive learning [13], [14] or adversarial learning [15], yet can achieve promising unsupervised time series anomaly detection performance simply by a fully connected network architecture. ALAD is built upon the core of activation learning [16], a new learning paradigm we have recently proposed for neural networks training. Quite different from BP, activation learning relies on an improved Hebbian learning rules [17] to induce local competition among neurons for network training, thus it does not require predefining a task-specific loss function or any feedback of error signals. As a result, it can not only facilitate eliminating the burden of designing complicated anomaly-aware learning objectives, but also achieve neural networks training with biological plausibility and computational efficiencies. More importantly, in activation learning, common input patterns can be learned more and activated more. The output activation value can reflect the relative typicality of the data in the whole data set. In time series anomaly detection, the aim is to mine anomalous temporal patterns, i.e., patterns that rarely or never occur. Correspondingly, the mechanism of activation learning facilitates the intuitive understanding of anomalies based on the activation values, thus it motivates us to apply activation learning to time series anomaly detection.

The proposed method shows advantages from both theoretical and practical perspectives. Theoretically, it does not rely on BP or task-specific loss functions and provides an intuitive and effective way to interpret detected anomalies based on activation values, offering a solution for designing novel anomaly detection models. Practically, our approach establishes criteria for identifying anomalies based on statistical distribution, avoiding the ambiguity of manually defined anomalies. Additionally, our method employs a simple network architecture and updates parameters via forward propagation, requiring less storage for intermediate results, making it more efficient and suitable for various practical applications. Overall, the main contributions of this work can be summarized as follows:

- A new unsupervised time series anomaly detection paradigm based on activation learning is proposed. This

is the first biologically plausible algorithm trained neural network for time series anomaly detection to the best of our knowledge.

- ALAD does not rely on BP training, complicated network structures, or tailored anomaly-aware learning objectives. A simple fully connected network architecture and local learning rules are all that is needed to achieve competitive anomaly detection performance.
- By utilizing various synthetic and real-world time series anomaly datasets, experimental results have illustrated the effectiveness and feasibility of the proposed paradigm. More importantly, this work has also demonstrated that biologically plausible algorithm trained models can outperform BP in certain scenarios.

The rest of this paper is organized as follows. Section II presents the related work, including the anomaly detection in time series and Hebbian learning. Section III covers the preliminary work. Section IV introduces the framework of the proposed method. The experimental results and discussion are given in Section V, followed by the conclusion and future work.

II. RELATED WORK

A. Anomaly Detection in Time Series

Anomaly detection in time series is an emerging topic in data mining and machine learning communities. Various anomaly detection methods have been proposed and shown their superiority in some specific situations, which mainly can be divided into statistical-based methods, machine learning-based methods and deep learning-based methods.

1) *Statistical-based Methods*: Statistical-based methods generally utilize traditional statistical algorithms to identify anomalies. For example, Auto-Regressive Integrated Moving Average (ARIMA) is one of the widely used tools to analyze time series data, a series of ARIMA-based methods have been proposed to exploit the temporal pattern of time series data and perform anomaly detection [18]. Additionally, methods that rely on statistical distribution assumptions and those based on distance measurements are also prevalent, offering different approaches to identifying deviations from normal patterns [19], [20]. However, real-world time series data generally exhibit volatile characteristics due to various internal/external factors, and these classical methods intrinsically lack the capability of capturing the temporal dependencies among time series data [21].

2) *Machine Learning-based Methods*: Machine learning-based methods apply modern techniques to detect anomalies. For instance, Isolation Forest (IF) [22] relies on an ensemble of isolation trees, which recursively randomly partitions the dataset to find isolated anomalies. Local Outlier Factor (LOF) [23] is a density-based anomaly detection algorithm, which performs anomaly detection by judging on the density between data points with their neighbors. Ji et al. [24] developed a hybrid model that projected multivariate time series into a lower-dimensional space to facilitate anomaly detection. Hu et al. [25] introduced a computational framework for identifying discords in multivariate time series using a recurrence plot analysis, alongside a strategy for efficient pairwise distance comparisons.

3) *Deep Learning-based Methods*: Nowadays, a series of studies have demonstrated the effectiveness of deep learning-based methods in time series anomaly detection [26]. Generally, deep learning-based methods are divided into two categories: forecasting-based and reconstruction-based methods [27]. Forecasting-based methods perform anomaly detection based on deviations between predicted results and actual values, typically using models based on LSTM [28], TCN [7], and Transformer [29]. Reconstruction-based methods typically employ autoencoders (AE) [30], variational autoencoders (VAE) [31], and their variants [32] to learn latent representations from normal data. These methods conduct anomaly detection by comparing reconstruction errors between the reconstructed and actual samples, operating under the assumption that the models would fail to accurately reconstruct unseen abnormal samples. However, these assumptions do not always hold. For instance, these models often rely on complicated architectures and focus predominantly on reconstruction or prediction quality, potentially causing even abnormal data to be well reconstructed or fitted due to ignored inconsistencies in latent representations. Moreover, the fundamental algorithm for training these deep learning-based methods is BP, which not only requires predefined loss functions for error propagation but also demands additional resources for gradient computations and storage. Overall, there remains an open need for designing more innovative solutions to expand the application scenarios of time series anomaly detection.

B. Unsupervised Anomaly Detection Methods

In the landscape of anomaly detection, supervised and unsupervised methods are widely utilized. Supervised anomaly detection relies on labeled datasets to train models that can distinguish between normal and anomalous patterns. In contrast, unsupervised anomaly detection methods do not require labeled data, focusing instead on learning the normal patterns from the data and identifying deviations as anomalies. This lack of dependence on labeled data makes unsupervised methods particularly significant, as they are more adaptable to a variety of real-world applications where labeling is impractical or impossible [33].

Unsupervised anomaly detection methods generally leverage various architectures, including autoencoders, graph neural networks, and Transformers, to detect anomalies by learning normal data patterns and identifying deviations without the need for labeled data [34]. For example, Unsupervised Anomaly Detection (USAD) [35] adopts an autoencoder architecture within an adversarial training framework, which amplifies the reconstruction error of anomalies and contributes to anomaly detection. Graph Deviation Network (GDN) [8] combines a structure learning module with a graph neural network and uses forecasting deviations as anomaly scores. TranAD [36], a state-of-the-art anomaly detection method, is based on the Transformer architecture and uses a Transformer encoder and self-conditioning mechanism to achieve representation learning and perform anomaly detection. OmniAnomaly [37] learns robust representations from time series using a stochastic recurrent neural network, reconstructs the input data from these representations, and uses the

reconstruction probabilities to identify anomalies. Li et al. [38] introduced a prototype-oriented unsupervised anomaly detection method for multivariate time series, enhancing adaptability through a probabilistic framework that utilizes transferable prototypes. These methods amplify reconstruction errors or use robust representations to identify anomalies, showcasing the adaptability of unsupervised approaches in handling complex datasets.

Compared to existing unsupervised methods, ALAD demonstrates several advantages. It leverages activation learning to utilize the global information of the dataset for estimating statistical aspects of time series patterns, offering an intuitive interpretation of anomalies based on activation values. This approach eliminates the need for predefined task-specific loss functions and reduces resource usage by avoiding gradient computations and storage. Furthermore, by not relying on backpropagation, ALAD aligns more closely with biologically plausible models of learning, providing a more efficient and scalable alternative in real-world settings.

C. Hebbian Learning

Hebbian learning is a biologically plausible and ecologically valid learning mechanism, which follows the principle of ‘units that fire together, wire together’ [39]. In the literature, there are several researchers have attempted to develop Hebbian-like mechanisms for training neural networks and as an alternative to BP, since it can address the biological implausibility of BP while not require any feedback of error signal, thus mitigated the computational inefficiencies in BP. Hebbian learning is used for neural network training by changing the strength of the synapses between neurons, i.e., the connection weights. Formally, given a synapse of weight w_{ij} connecting neuron a_i in layer l and neuron a_j in layer $l + 1$, the rule of vanilla Hebbian learning can be defined by

$$\Delta w_{ij}^l = \eta a_i^l a_j^{l+1}, \quad (1)$$

where a_i^l and a_j^{l+1} denote the activation values of pre-synaptic neuron and post-synaptic neuron, respectively, and η is the learning rate. The core idea of Hebbian learning is that connection weight w_{ij} should be associated with both pre-synaptic and post-synaptic neurons. When both of the two neurons are activated at the same time, the connection weight between them should be strengthened, while when the two neurons are in opposite states, the connection weight should be weakened.

Recently, motivated by Hebbian learning rules, Krotov and Hopfield [40] proposed a learning algorithm for training neural network, which leveraged the global inhibition in hidden layers and was capable of learning lower-layer early representations that facilitated the training of high-layer weights with a regular backpropagation algorithm on various tasks. HebbNet [41] was an improved Hebbian learning based neural network, which applied an activation threshold and gradient sparsity into the Hebbian learning rules. Similar to the training procedure designed by Krotov and Hopfield, the first layer weights in HebbNet were learnt in an unsupervised manner by the proposed Hebbian learning rules, and the weights between hidden and

output layer were learnt by BP. HebbNet achieved comparable performance to Krotov and Hopfield, while simplifying the training procedure. SoftHebb [42] combined a soft winner-takes-all strategy with Hebbian learning rule to minimize cross-entropy, which was proved to be capable of increasing learning speed and higher robustness to noise and adversarial attacks. Grounded on SoftHebb, Adrien et al. [43] presented multilayer SoftHebb, which achieved depth in Hebbian learning and was verified to realize promising performance on difficult tasks with relatively high efficiency.

However, there are very few attempts to further improve the effectiveness of Hebbian learning in training neural networks. Besides, existing approaches only adopt Hebbian learning-based methods to solve image classification tasks while not applying Hebbian learning to other specific scenarios. Thus, the potential of Hebbian learning-based methods has been underestimated. In this work, we proposed a Hebbian learning-improved method for time series anomaly detection, which broadens the applications scenarios of Hebbian learning algorithms and also demonstrates that biologically plausible algorithm trained models can outperform BP in certain scenarios.

III. PRELIMINARIES

A. Activation Learning Rules

In the vanilla Hebbian learning, the rule in (1) is not stable since the weights may grow unboundedly. A series of modified rules have been proposed to improve Hebbian learning in terms of convergence, which typically introduce additional constraints or restrictions [17], [44], but this reduces the biological plausibility. Moreover, such rules are used to train a single neuron and are not applicable to the training of multiple neurons in a layer. Therefore, by introducing a competition mechanism within neurons and among neurons into Hebbian learning, we develop the activation learning rule in [16], which can work with the training of multiple neurons in a layer:

$$\Delta w_{ij}^l = \eta a_j^{l+1} \left(a_i^l - \sum_k a_k^{l+1} w_{ik}^l \right). \quad (2)$$

In the learning rule, $a_j^{l+1} = \sum_r a_r^l w_{rj}^l$ denotes the result of feedforward propagation to neuron j from the previous layer. The term $\sum_k a_k^{l+1} w_{ik}^l$ represents the internal feedback from all the neurons in layer $l+1$, which forces the neurons in layer $l+1$ to compete with neuron j to fire. To understand this phenomenon, assuming that a_j^{l+1} , a_k^{l+1} and w_{ik}^l are all positive variables, and if a_k^{l+1} is activated, then the connection weight w_{ij}^{l+1} is weakened by a strength of $\eta a_j^{l+1} a_k^{l+1} w_{ik}^l$. After iterative training, certain weights tend to activate more for specific input patterns. This mechanism we call local competition, since it raises local competitions within and among neurons.

The weight matrix $\mathbf{w}$ of each layer can converge to an asymptotically stable solution after a sufficient number of training steps by learning rule (2), which takes the form of

$$C\mathbf{w} - \mathbf{w}\mathbf{w}^T C\mathbf{w} = 0, \quad (3)$$

where $C = E\{\mathbf{x}\mathbf{x}^T\}$ is the covariance matrix of inputs. For the number of receiving neurons $m = 1$, there are two stable solutions of $\mathbf{w}$, corresponding to $\mathbf{v}$ and $-\mathbf{v}$, where $\mathbf{v}$ is the eigenvector corresponding to the largest eigenvalue of C . For $m > 1$, there are infinite number of stable solutions, and the specific stable solution depends on the initial value of weight and the training samples.

B. Problem Statement

Given a time series data $\mathcal{S} = \{x_1, x_2, \dots, x_T\}$ with length T , each point x_t is collected at a specific timestamp t . For time series anomaly detection, historical data are beneficial to understand the contextual relationships of an observation x_t . Therefore, like most methods [35], [37], a lookback window with size n is applied to convert time series data into a sequences of sliding windows, and each window is used to calculate the anomaly result, rather than just x_t . At a time step t , the input time series data in window $\mathbf{X}_t$ can be represented as $\mathbf{X}_t = [x_{t-n+1}, x_{t-n+2}, \dots, x_t]^T$. The task of time series anomaly detection is to assign a binary label $y_t \in \{0, 1\}$ to indicate whether timestamp t is anomalous or not.

IV. METHOD

In this section, we first introduce how activation values are utilized for anomaly detection. Subsequently, we detail the network architecture that we have adopted. Finally, we describe the manner of training and inference.

A. Activation Value for Input Typicality Estimation

Given a layer trained by the above learning rule (2), if the input strength of $\|\mathbf{a}^l\|^2$ is fixed, a more common input pattern will induce a higher activation $\|\mathbf{a}^{l+1}\|^2$, which means that the output strength of activation can be used to estimate the typicality of the input pattern, i.e., how probable the input pattern is. To further explain the phenomenon, assume that the number of receiving neurons in next layer m is smaller than the input dimension and V represents the vector space spanned from the eigenvectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_m$. Wherein, V is called typical space, which is a convex and compact area shaped by multiple layers, and it contains majority of the common inputs, thus the distance between it and an input sample can be used to measure the typicality of the input. Fig. 1 visually demonstrates how the output activation values are used to measure the typicality of the time series patterns.

Given the input vector of a layer $\mathbf{a}^l$, the distance between it and the typical space can be expressed as

$$d(\mathbf{a}^l, V) = \sqrt{\|\mathbf{a}^l\|^2 - \sum_i \|\mathbf{a}^{lT} \mathbf{v}_i\|^2}. \quad (4)$$

More importantly, from the stable solutions described in [16], it is derived that the squared norm of the output activation, $\|\mathbf{a}^{l+1}\|^2$, tends to be upper bounded by

$$\|\mathbf{a}^l\|^2 - d^2(\mathbf{a}^l, V). \quad (5)$$

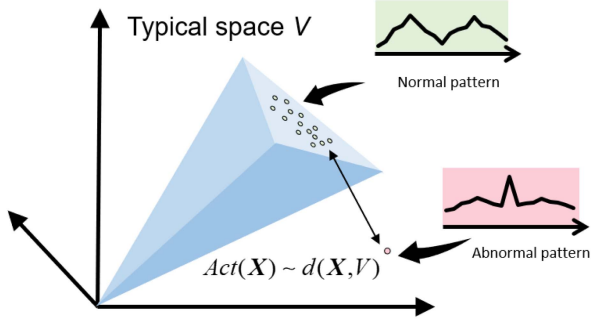


Fig. 1. The output activation measures the typicality of the time series pattern, and anomalous patterns will be far from the typical space, leading to smaller activation values.

In particular, when $\mathbf{w} = [\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_m]$, $\|\mathbf{a}^{l+1}\|^2$ converges to $\|\mathbf{a}^l\|^2 - d^2(\mathbf{a}^l, V)$. Consequently, when the magnitude of the input $\|\mathbf{a}^l\|^2$ is fixed, the output activation $\|\mathbf{a}^{l+1}\|^2$ can be used to measure the distance between the input $\mathbf{a}^l$ and the typical space V . It should be noted that although only the case that m is less than the input dimension is discussed, the above conclusion still holds for generic m .

Overall, the aforementioned useful insights in activation learning motivate us to apply it to time series anomaly detection. Specifically, in realistic time series data, anomalous patterns generally rarely or never occur, which causes difficulties in establishing anomaly rules manually. Fortunately, the mechanism of activation learning is capable of leveraging the global information of the dataset to estimate the statistical aspects of the time series patterns, and it further provides an intuitive way to interpret the detected anomalies based on the activation values.

B. Network Architecture

In this subsection, the designed network architecture is presented. To incorporate the mechanism of activation learning, we design a simple but tailored architecture based on a fully connected network, as shown in Fig. 2.

Specifically, the original time series data are first transformed into sliding windows for keeping the contextual relationships, and then fed into the ALAD. In the bottom-up network structure of ALAD, the sliding windows are first fed to a norm layer, which transforms the input data into a unit vector to fix the input magnitude. Then, following is a fully connected network with multiple hidden layers. In order to cascade multiple layers in a network, the input and output magnitudes of each layer need to be maintained consistent to control the activation output, i.e., $\|\mathbf{x}\|_2 = \|f(\mathbf{x})\|_2$. Therefore, the following form of nonlinear function is adopted here as the activation function

$$f(\mathbf{x}) = \beta(|\mathbf{x}| - \overline{|\mathbf{x}|}), \quad (6)$$

which serves to de-mean the absolute value of the input $\mathbf{x}$, and β is a coefficient used to rescale $\|\mathbf{x}\|_2 = \|f(\mathbf{x})\|_2$. It should be noted that any nonlinear function that does not change the magnitude of the input can be considered as a candidate for the activation function.

As mentioned previously, the output of a layer trained by the learning rule (2) is prone to generating higher activation value on a more probable pattern. In order to output a scalar value for evaluating directly whether the input pattern is abnormal or not, an output activation layer is deployed on the top of ALAD, which performs $\|\cdot\|_2^2$ operation on the output of the last hidden layer.

By employing multiple layers with specific nonlinear activation functions, ALAD constructs a compact typical space that effectively captures the probability distribution of input patterns. In time series anomaly detection, this is crucial because anomalous patterns often have a low probability of occurrence or may not appear in the training dataset at all. During the training phase, this approach ensures that anomalous patterns are situated far from the typical space, resulting in smaller activation values. These values serve as a critical metric for identifying anomalies, as they signify substantial deviations from expected patterns.

ALAD enhances anomaly detection by leveraging activation learning to effectively utilize global dataset information, facilitating intuitive interpretations of anomalies through activation values and statistical insights, thereby alleviating the ambiguity in defining ground-truth anomalies. The method employs a simple network architecture, consisting of just a few fully connected layers, and it does not require the definition of a global task-specific loss function, thereby minimizing resource demands such as gradient computations and storage. Additionally, its simplified structure also enables potential integrations with advanced deep learning models, offering opportunities for model initialization and combining with pretrained models.

C. Model Training

Activation learning adopts local Hebbian learning rules to update the synaptic weights. Therefore, the training of ALAD with a multi-layer network structure is conducted in a bottom-up unsupervised manner. The training process of ALAD is summarized in Algorithm 1. Specifically, the weight matrix of the network is first initialized from a normal distribution with zero mean and small variance. Then, input data is normalized to unit vector before being fed into the network. Subsequently, forward propagation is performed and the weight matrixes are trained layer by layer by learning rule. This process is repeated until the specified epochs.

The above training method can also be applied to batch-wise training manner. Specifically, in each training epoch, the training samples are divided into multiple mini-batches fed into the network for training the weights through feedforward and local feedback, such that the weights of the multi-layer network can be updated layer by layer with each batch. Given a batch of samples, the learning rule (2) can be modified as

$$\Delta w_{ij}^l = \frac{\eta}{B} \sum_{b=1}^B a_{j(b)}^{l+1} \left(a_{i(b)}^l - \sum_k a_{k(b)}^{l+1} w_{ik}^l \right), \quad (7)$$

where B denotes the batch size, b represents the index of samples in a batch, and η is the learning rate.

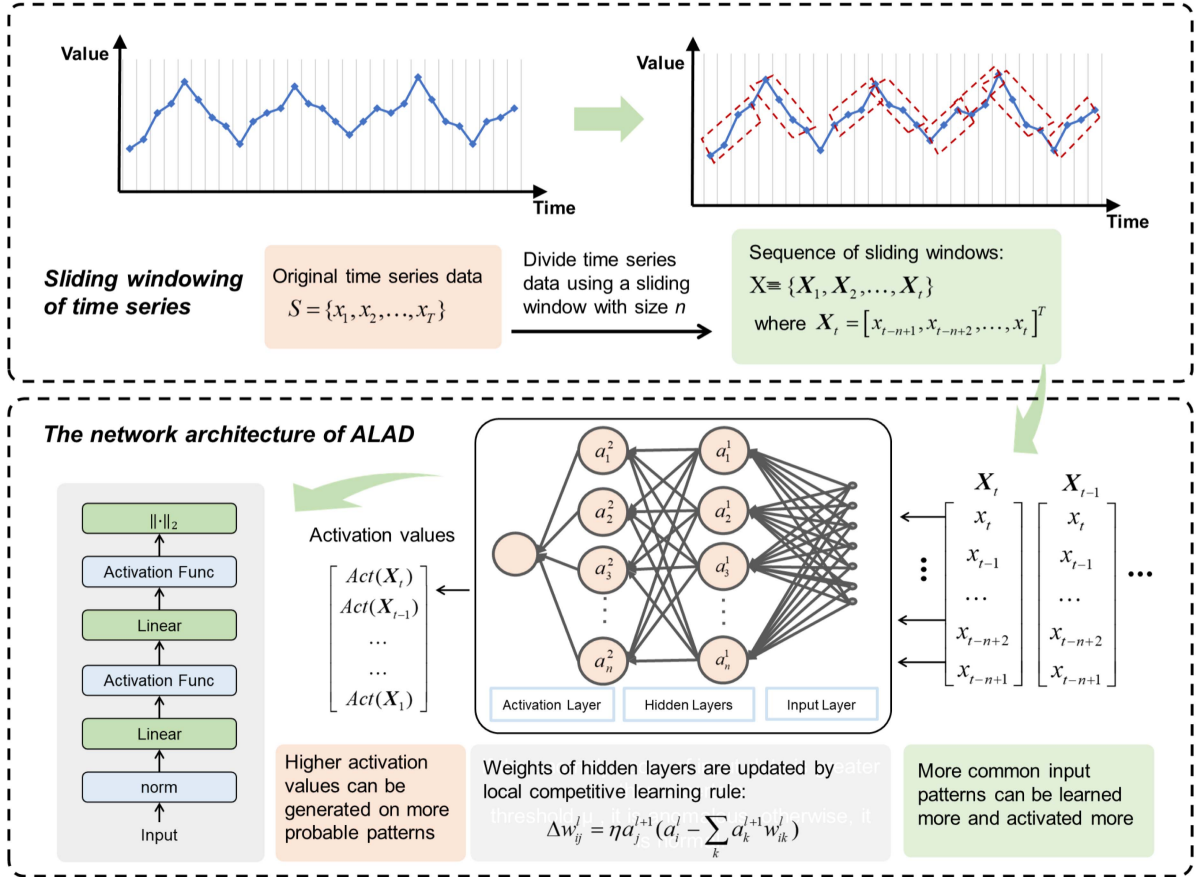


Fig. 2. The framework of ALAD. The upper subfigure of the framework shows the sliding windowing of the original time series. The lower subfigure shows how ALAD works and the network architecture details.

D. Anomaly Inference

The inference phase of ALAD is summarized in Algorithm 2. During the inference phase, the data $\mathbf{X}_t$ in the sliding window is fed into the trained model, and then being normalized to a unit vector. Then, the output of the last hidden layer is obtained via layer by layer forward propagation. Subsequently, the activation value of $\mathbf{X}_t$ is derived by $\|\cdot\|_2^2$ operation. In the mechanism of activation learning, the more common patterns induce larger activation values, while the rarer patterns induce smaller activation values, thus for intuitiveness, $1 - Act(\mathbf{X}_t)$ is used to indicate the anomaly score of input data. If the anomaly score of input data is greater than a certain threshold μ , it is anomalous, otherwise, it is normal. In this work, similar to [22], [23], the threshold value is obtained through grid search. Notably, our model's output activation values reflect the distribution of data patterns in the training set, enabling anomaly detection by identifying anomalously low activation values during testing. Thus, in practical scenarios, even in the absence of anomaly samples, we can determine the threshold μ by setting it at various percentiles within normal data, such as the 95th percentile. This threshold can be further adjusted based on the specific definition of anomalies or prior knowledge relevant to the application context.

V. EXPERIMENTS

A. Datasets

We use synthetic datasets and real-world datasets for the performance evaluation of all the methods. The descriptions of these datasets are as follows:

Synthetic Datasets: We apply the synthesizing strategy from [45] to inject anomalies into time series, which provide a general and unified synthetic criterion for generating various anomalies from different applications. Wherein, 24 synthetic univariate time series data are generated, which contain six types of anomalies covering point-wise and pattern-wise anomalies, including global, contextual, shapelet, seasonal, trend and mixture anomalies. Specifically, the synthetic univariate time series are generated with sinusoidal wave and each series has a length of 2000. The first half of the data is used as training data and the rest is used as test data with different proportions of anomalies injected. Each type of anomalies is manipulated with different ratio 0.05, 0.1, 0.15 and 0.2.

Real-World Datasets: We use four publicly available datasets collected from real-world application scenarios. These include CPU usage data from an Amazon server (Service), real-time traffic data from the Twin Cities Metro area in Minnesota (Traffic) [46], internal bleeding data derived from arterial blood

Algorithm 1: Algorithm for Training ALAD.

Input: The set of training data $\mathcal{X} = \{\mathbf{X}_1, \mathbf{X}_2, \dots, \mathbf{X}_T\}$, learning rate η , epochs E .

Output: L layers of trained weight matrixes $\mathbf{W}_1^*, \mathbf{W}_2^*, \dots, \mathbf{W}_L^*$.

- 1: Random initialize L layers of weight matrixes $\mathbf{W}_1, \mathbf{W}_2, \dots, \mathbf{W}_L$ from a normal distribution of zero mean and small variance.
- 2: **while** $e < E$ **do**
- 3: **for all** $t = 1, 2, \dots, T$ **do**
- 4: //normalize input
- 5: $\mathbf{A}_0 \leftarrow \text{Norm}(\mathbf{X}_t)$
- 6: **for all** $l = 1, \dots, L$ **do**
- 7: //forward pass
- 8: $\mathbf{A}_l = \mathbf{W}_l^T \mathbf{A}_{l-1}$
- 9: //use learning rule to train weights layer by layer
- 10: $\Delta \mathbf{W}_l \leftarrow (\mathbf{A}_{l-1} - \mathbf{W}_l \mathbf{A}_l) \mathbf{A}_l^T$
- 11: $\mathbf{W}_l \leftarrow \mathbf{W}_l - \eta \Delta \mathbf{W}_l$
- 12: $\mathbf{A}_l \leftarrow f(\mathbf{A}_l)$
- 13: **end for**
- 14: **end for**
- 15: $e \leftarrow e + 1$
- 16: **end while**
- 17: **return** $\mathbf{W}_1^*, \mathbf{W}_2^*, \dots, \mathbf{W}_L^*$.

pressure measurements in pigs (Bleeding) [47], and a subset of the UCR dataset, which includes data from various natural sources [48]. The characteristics of real-world datasets are summarized in Table II.

B. Experimental Settings

Comparison Methods. We make a comparative study of the proposed method with nine representative methods, including IF [22], LOF [23], AE [30], USAD [35], DAGMM [49], GDN [8], TranAD [36], DCdetector [48] and Anomaly-BERT [50]. For each comparison method, we use grid search to select anomaly thresholds for different datasets separately.

Datasets Configuration. To maintain consistency, all methods use a sliding window of size 5, with overlapping windows for the real dataset and non-overlapping windows for the synthetic dataset. For the real-world dataset, we use the same dataset division proportion as [36]. For the synthetic dataset, all methods are trained on the unmanipulated data and tested on the manipulated data.

Evaluation Metrics. To measure and evaluate the performance of the proposed method, similar to previous studies [36], three common metrics, Recall, Precision and F1 score are employed to compare the performance of different models, these metrics are derived:

$$\text{Precision} = \frac{TP}{TP + FP} \quad (8)$$

$$\text{Recall} = \frac{TP}{TP + FN} \quad (9)$$

Algorithm 2: Algorithm for Anomaly Inference.

Input: The set of test data $\tilde{\mathcal{X}} = \{\mathbf{X}_1, \mathbf{X}_2, \dots, \mathbf{X}_T\}$, threshold μ .

Output: The label y_t of $\mathbf{X}_t$.

- 1: Load the trained model with weight matrixes $\mathbf{W}_1^*, \mathbf{W}_2^*, \dots, \mathbf{W}_L^*$.
- 2: **for all** $t = 1, 2, \dots, T$ **do**
- 3: //normalize input
- 4: $\mathbf{A}_0 \leftarrow \text{Norm}(\mathbf{X}_t)$
- 5: **for all** $l = 1, \dots, L$ **do**
- 6: //forward pass
- 7: $\mathbf{A}_l = f(\mathbf{W}_l^{*T} \mathbf{A}_{l-1})$
- 8: **end for**
- 9: //calculate the output activation of $\mathbf{X}_t$
- 10: $\text{Act}(\mathbf{X}_t) \leftarrow \|\mathbf{A}_L\|_2^2$
- 11: **if** $(1 - \text{Act}(\mathbf{X}_t)) > \mu$ **then**
- 12: $y_t = \text{"Anomalous"}$
- 13: **else**
- 14: $y_t = \text{"Normal"}$
- 15: **end if**
- 16: **end for**

$$F1 \text{ score} = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \quad (10)$$

where TP is the True Positives, FP is the False Positives, and FN is the False Negatives. In actual cases, anomalies typically last for a period of time and thus form a continuous segment of anomalies. Therefore, we do not care about point-by-point evaluation, but use a point-adjust evaluation approach [51], which is widely used in the evaluation of anomaly detection tasks [36], [52]. Specifically, for a continuous segment of anomalies, if one of the observations from the ground truth is correctly identified, then all anomalies in the segment are viewed to be correctly detected, and observations outside the ground truth anomaly segment are treated as normal.

C. Experimental Results on Real-World Datasets

In this subsection, experiments on real-world datasets are conducted to evaluate the performance of ALAD. The experimental results are shown in Table I, where three metrics are calculated, and P, R, F1 respectively represent the Precision, Recall, F1 score. For these metrics, a higher value indicates a better performance. Specifically, the best F1 scores are bolded, and in particular, the average F1 scores are also provided for the Traffic, Bleeding and UCR datasets.

From the experimental results, one can find that ALAD has achieved competitive performance than comparison methods on most datasets. In particularly, ALAD has realized the best average F1 scores on Bleeding, Traffic and UCR datasets. Machine learning-based methods, like IF and LOF achieve worst performance, since they are both unsupervised methods that consider each instance as an individual, thus they are susceptible to noisy and perturbed data. USAD, DAGMM and TranAD are recent representative reconstruction-based methods, they

TABLE I
EXPERIMENTAL RESULTS FOR ALAD WITH COMPARISON METHODS ON REAL-WORLD DATASETS

Method	Service			Traffic_1			Traffic_2			Traffic_3			Traffic_Avg
	P	R	F1	P	R	F1	P	R	F1	P	R	F1	F1
IF	0.210	0.999	0.347	0.246	0.499	0.329	0.136	0.999	0.240	0.046	0.999	0.088	0.219
LOF	0.521	0.999	0.685	0.181	0.999	0.307	0.046	0.999	0.088	0.190	0.999	0.319	0.238
AE	0.333	0.999	0.499	0.410	0.499	0.450	0.640	0.999	0.780	0.347	0.999	0.516	0.582
USAD	0.842	0.666	0.744	0.640	0.666	0.653	0.571	0.999	0.727	0.727	0.999	0.842	0.740
DAGMM	0.857	0.999	0.923	0.680	0.999	0.810	0.551	0.999	0.711	0.888	0.999	0.941	0.820
GDN	0.888	1.000	0.941	0.680	0.999	0.810	0.571	0.999	0.727	0.615	0.999	0.761	0.766
TranAD	0.923	0.999	0.959	0.888	0.749	0.813	0.727	0.999	0.841	0.657	0.999	0.792	0.815
DCdetector	0.962	0.999	0.980	0.790	0.999	0.882	0.938	0.999	0.967	0.999	0.999	0.999	0.949
AnomalyBERT	0.958	0.999	0.978	0.860	0.999	0.924	0.943	0.999	0.970	0.999	0.999	0.999	0.964
ALAD	0.879	1.000	0.935	0.888	0.999	0.940	0.941	0.999	0.969	0.999	0.999	0.999	0.969

Method	Bleeding_1			Bleeding_2			Bleeding_3			Bleeding_4			Bleeding_Avg
	P	R	F1	P	R	F1	P	R	F1	P	R	F1	F1
IF	0.004	0.999	0.008	0.041	0.999	0.079	0.044	0.999	0.084	0.005	0.999	0.010	0.045
LOF	0.031	0.999	0.061	0.511	0.999	0.676	0.645	0.999	0.784	0.181	0.999	0.307	0.457
AE	0.074	0.999	0.139	0.668	0.999	0.801	0.221	0.999	0.362	0.666	0.999	0.799	0.525
USAD	0.749	0.999	0.857	0.616	1.000	0.762	0.684	1.000	0.812	0.714	0.999	0.833	0.816
DAGMM	0.363	0.999	0.533	0.676	1.000	0.807	0.739	1.000	0.849	0.666	0.999	0.799	0.747
GDN	0.888	1.000	0.941	0.672	1.000	0.804	0.739	1.000	0.849	0.999	0.999	0.999	0.898
TranAD	0.796	0.999	0.886	0.860	1.000	0.924	0.725	0.999	0.840	0.940	0.999	0.968	0.905
DCdetector	0.832	0.999	0.908	0.918	0.999	0.957	0.999	0.999	0.999	0.903	0.999	0.949	0.953
AnomalyBERT	0.812	0.999	0.896	0.902	0.999	0.947	0.999	0.999	0.999	0.896	0.999	0.945	0.947
ALAD	0.845	0.999	0.916	0.932	1.000	0.965	1.000	1.000	0.999	0.909	0.999	0.952	0.958

Method	UCR_1			UCR_2			UCR_3			UCR_4			UCR_Avg
	P	R	F1	P	R	F1	P	R	F1	P	R	F1	F1
IF	0.643	0.562	0.600	0.571	0.499	0.533	0.545	0.429	0.480	0.273	0.499	0.353	0.492
LOF	0.375	0.188	0.250	0.571	0.499	0.533	0.556	0.714	0.625	0.333	0.182	0.235	0.411
AE	0.696	0.999	0.821	0.500	0.999	0.667	0.455	0.357	0.400	0.333	0.364	0.348	0.559
USAD	0.625	0.999	0.769	0.714	0.999	0.833	0.565	0.999	0.722	0.769	0.999	0.869	0.798
DAGMM	0.600	0.999	0.750	0.650	0.999	0.788	0.620	0.999	0.765	0.630	0.999	0.773	0.769
GDN	0.550	0.999	0.709	0.570	0.999	0.726	0.590	0.999	0.742	0.580	0.999	0.734	0.728
TranAD	0.890	0.999	0.941	0.940	0.999	0.969	0.870	0.999	0.930	0.950	0.999	0.974	0.953
DCdetector	0.900	0.999	0.947	0.920	0.999	0.958	0.940	0.999	0.969	0.960	0.999	0.979	0.963
AnomalyBERT	0.915	0.999	0.955	0.935	0.999	0.966	0.955	0.999	0.977	0.975	0.999	0.987	0.971
ALAD	0.955	1.000	0.977	0.967	0.925	0.978	0.981	1.000	0.991	0.978	1.000	0.989	0.984

TABLE II
REAL-WORLD DATASET STATISTICS

Dataset	Training	Test	Anomaly(%)
Service	4032	4032	0.59
Traffic_1	1127	1127	2.83
Traffic_2	2495	2495	0.64
Traffic_3	2500	2500	0.32
Bleeding_1	1200	6301	0.19
Bleeding_2	1600	5900	1.80
Bleeding_3	2300	5200	1.96
Bleeding_4	3000	4500	0.22
UCR_1	3200	4301	0.56
UCR_2	1200	6301	0.39
UCR_3	3000	4500	0.56
UCR_4	4200	3301	0.59

only use local window as input for reconstruction, and cannot fully utilize the global information of the entire dataset. Moreover, reconstruction-based methods mainly focus on the quality of instance reconstruction, thus even abnormal data could be well reconstructed. Forecasting-based methods, like GDN, tries to predict the values based on historical data, and performs anomaly detection according to the forecasting errors. However, real-world time series data are inherently unpredictable due

to various internal/external factors. Thus, normal observations would be easily mistaken detected by GDN and cause a false positive. DCdetector and AnomalyBERT are state-of-the-art methods that have achieved commendable results, but DCdetector's reliance on dual attention contrastive learning may not adapt well to datasets where anomaly signatures are subtle and highly variable, leading to less effective anomaly discrimination. Similarly, AnomalyBERT's dependency on self-supervised learning with predefined synthetic outliers can limit its ability to generalize across real-world datasets that exhibit highly diverse and unforeseen anomaly patterns. More importantly, real-world time series data are generally composed of a large of number of short-term behaviors with various temporal patterns. The more common patterns can be learned more and activated more, while the anomalous patterns that exhibit distinct characteristics cannot be activated, thus, in ALAD the global information of dataset can be fully utilized.

In addition, from the experimental results, one can find that most methods achieve higher Recall but lower Precision. The reason behind this is that realistic anomalous patterns generally exhibit relatively distinct patterns, so that most methods can handle the detection of most anomalies, thus achieve high recall. However, time series data in realistic scenarios are

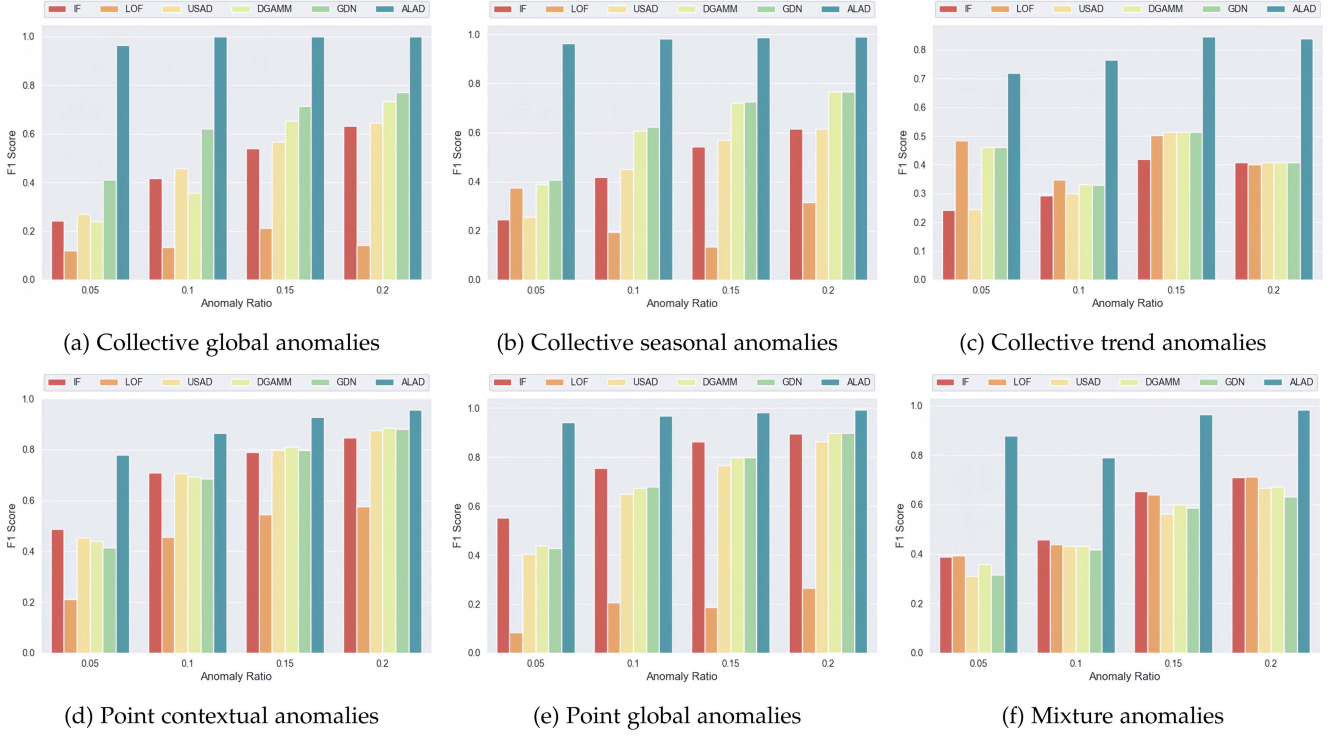


Fig. 3. Experimental results on synthetic datasets.

inevitably disturbed by high noises, uncertainties, etc. These realistic sophisticated compositions would considerably affect the performance of detection precision. Even so, ALAD still achieves competitive performance compared with the state-of-the-arts. It should be noted that ALAD does not use BP, meaning it does not rely on any anomaly-specific learning objectives, which makes the method more versatile for anomaly detection tasks. Additionally, ALAD employs a straightforward, fully connected network architecture without complex mechanisms, suggesting significant potential for further enhancements.

D. Experimental Results on Synthetic Datasets

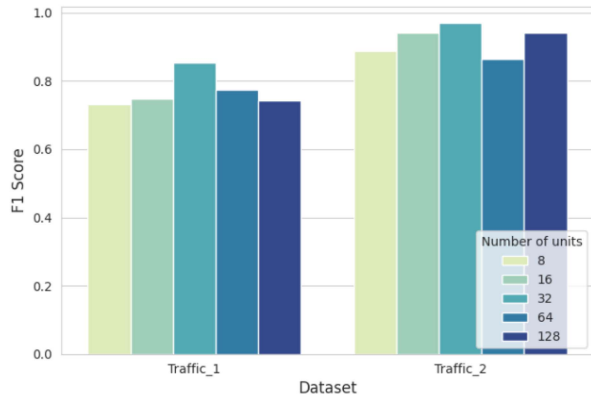
The experimental results on synthetic datasets are demonstrated in Fig. 3, where the F1 scores of each method in different ratios of anomalies are summarized. First, Fig. 3 demonstrates that ALAD achieves better results than other comparison methods. In particular, ALAD exhibits a competitive performance in detecting collective types of anomalies. To be specific, collective anomalies are defined as a collection of related data instances that is anomalous with respect to the entire dataset [45]. Accordingly, kinds of related data instances are fed into ALAD in the form of sliding windows, thus the contextual relationships can be fully utilized for activation learning. In this way, compared with the anomalies caused by individual points, anomalous patterns in a sequence of consecutive points are more likely to induce larger responses. In addition, it can also be found that collective trend anomalies are more challenging for ALAD. The reason behind this is that a norm layer is first applied in the structure of ALAD, which may damage the performance for identifying

trend anomalies when the window size is not appropriately set. Even though, ALAD still achieves better performance than the other methods. The main reason for the promising performance achieved by ALAD on synthetic data is that ALAD is trained on the training set without injected anomalies, which is facilitated to learn the normal patterns, thus the unseen abnormal patterns will be reflected more timely and accurately during the evaluation phase.

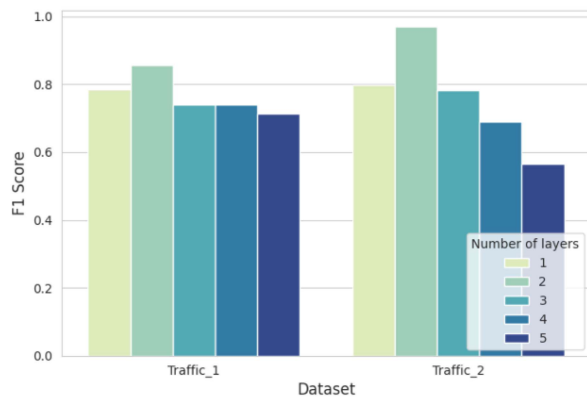
E. Case Study

To show the performance of ALAD in capturing the anomalous patterns of time series data intuitively, four cases of visualization of anomaly detection results are shown in Fig. 5. From the visualizations, one can see that ALAD achieves high precision and recall. ALAD aims at learning the statistical distribution of the input patterns, and it is capable of generating anomaly scores based on the rarity of patterns in training data, which means that ALAD does not require the hypothesis of pollution-free training set like existing reconstruction-based methods, thus it has more practical significance. Moreover, it can also be seen from the figures that ALAD raises different magnitudes of anomaly scores according to the anomalous degree of the input pattern, which is useful to distinguish the severity of anomalies in practical scenarios.

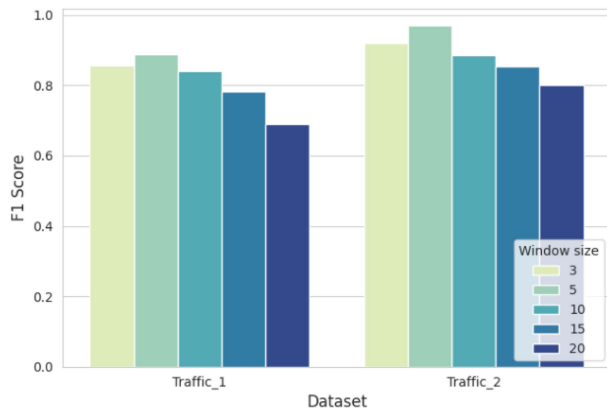
To visualize the statistical distribution of the samples learned by ALAD from an intuitive perspective, we add a hidden layer with 3 nodes after the last hidden layer in the network structure, which facilitates showing the distribution of different input patterns in 3D space. Specifically, ALAD is trained on the



(a)



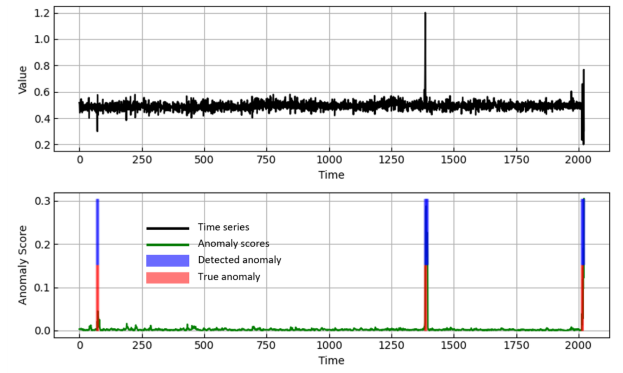
(b)



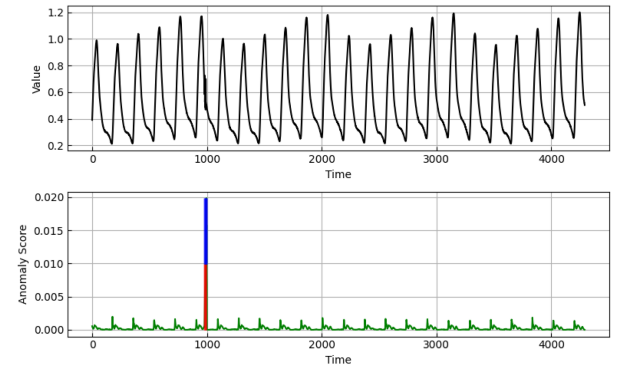
(c)

Fig. 4. Hyperparameters analysis for (a) the number of units, (b) the number of layers, and (c) the window size.

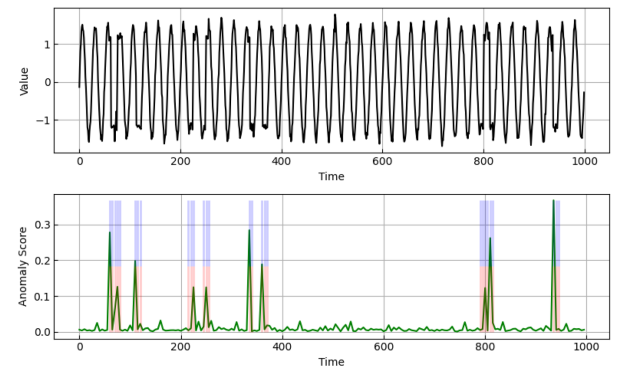
training set, then tested on a selected time series containing anomalous data and the results of the last hidden layer are visualized by projection into the 3D space. The results of the visualization on the collective global, point global, Bleeding_1 and Bleeding_2 datasets are shown in Fig. 6, where each black point denotes a normal sample and each red point denotes an abnormal sample. From the visualization results, it can be seen



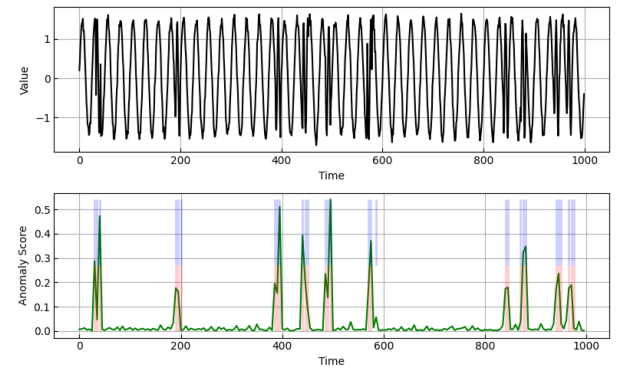
(a)



(b)



(c)



(d)

Fig. 5. Four cases of visualization of anomaly detection results on (a) service, (b) bleeding_1, (c) collective global anomalies, and (d) collective seasonal anomalies.

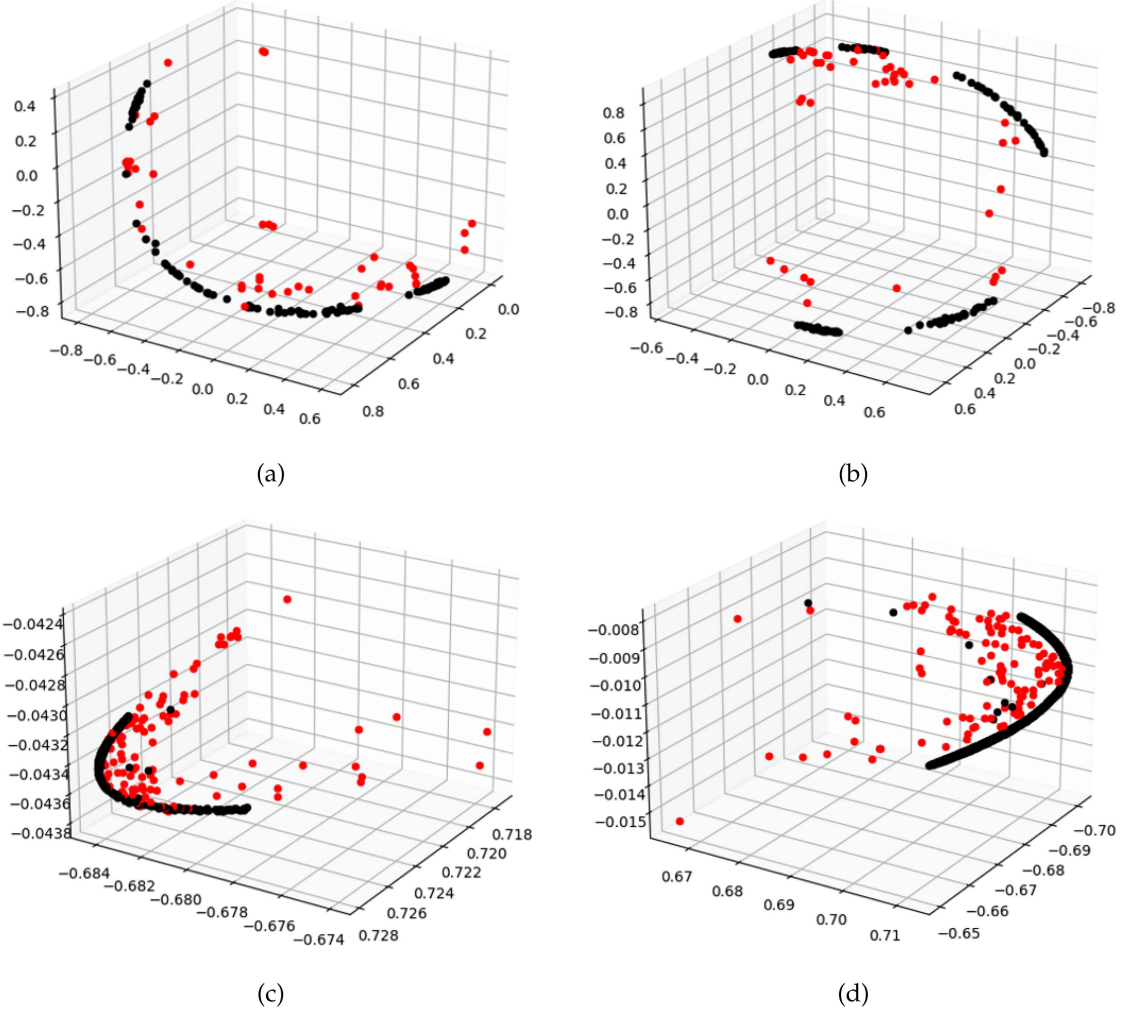


Fig. 6. Four cases of visualization of the learned statistical distribution results for (a) collective global anomalies, (b) point global anomalies, (c) bleeding_1, and (d) bleeding_2.

that the normal samples are concentrated on the arc, while the abnormal samples are scattered in the interior of the arc. The reason for this phenomenon is that, given a fixed strength of the input samples, normal samples usually have stronger activation values, i.e., L2 norm, causing them to be scattered on arcs farther from the center of the circle. The abnormal samples, in contrast, have lower activation values, so they are closer to the center of the arc. Consequently, the anomalous patterns are well distinguished from the normal patterns.

F. Analysis of Hyperparameters

The influences of three hyperparameters, i.e., the number of layers, the number of units per layer, and the window size, may impact performance. Therefore, the Traffic_1 and Traffic_2 datasets are used for analysis. Specifically, the number of units discussed ranges from $\{8, 16, 32, 64, 128\}$, the number of layers from 1 to 5, and the window sizes from $\{3, 5, 10, 15, 20\}$. When analyzing the effectiveness of one hyperparameter, the others

are kept unchanged. From the visualizations in Fig. 4, one can observe that different combinations have varying effects on the model's performance. Specifically, with an increase in hidden units given a fixed depth, the F1 score of ALAD first increases and then decreases. Similar observations can also be found in the analysis of the number of layers. This phenomenon can be attributed to the fact that smaller hidden units or fewer layers result in a simpler network structure, thus the model is incapable of learning effectively from data. Conversely, higher numbers of hidden units and layers may lead to a more complicated network structure, making the model susceptible to noise and harder to optimize. The discussion on window size indicates that different window sizes also have an impact on the experimental results. This occurs because if the window size is set too large, it becomes difficult to capture subtle anomalies, whereas a window size that is too small may not effectively detect anomalies that manifest over longer trends. Overall, since different time series data may exhibit different temporal characteristics, the combination of hyperparameters suitable for one dataset may not be appropriate

for another, thus it is necessary to choose hyperparameters based on empirical experience when applying ALAD to a specific dataset.

VI. CONCLUSION AND FUTURE WORK

In this work, based on activation learning, a new unsupervised time series anomaly detection paradigm is proposed, which provides an intuitive and effective way to interpret detected anomalies based on the activation values. Experimental results on various real-world and synthetic datasets confirm the promising performance of the proposed method compared with the state-of-the-art methods. Furthermore, this work has complemented the activity of existing Hebbian-like methods, and also demonstrated that biologically plausible algorithm can achieve more promising performance than BP in certain scenarios. In future work, we plan to extend the capabilities of our method to adaptive anomaly detection, for example by integrating peak detection techniques that can dynamically adjust based on the distribution of activation values, thereby enhancing its practical value in real-world scenarios. Additionally, as a new paradigm, our approach holds significant potential for integration with modern deep learning technologies such as pretrained models and self-supervised learning techniques. Moreover, we are particularly excited about the potential of activation learning to facilitate the design of innovative neural network architectures, broadening the application of our method across various domains.

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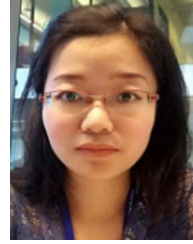
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